

Induction and recursion

Chapter 5

What is Induction?

- ▶ **INDUCTION IS A PROOF TECHNIQUE**
- ▶ Mathematical induction is a way to prove something is true for all numbers (1, 2, 3, ...).
- ▶ **Domino Example**
 - Suppose we align an infinite number of dominoes, such that if some domino falls, the next one in line falls as well. What happens when we knock down the first domino?
 - **So what happens?**
All the dominoes fall.
 - This intuition is formalized in the **principle of mathematical induction**.



How this relates to induction ?

To prove something for all numbers:

- ▶ **Start it (first domino):**
Show it works for $n = 1$
- ▶ **Keep it going (domino effect):**
Show that if it works for $n = k$, then it also works for $n = k + 1$



Why Induction ?

Induction is a useful method for:

- defining things,
- proving ideas,
- and writing programs.

It is **important in algorithms** because it helps us:

- prove that a program works correctly
- understand how fast it runs

Even though problems can be big and complex, induction helps by:

- breaking them into **small steps**
- solving one step at a time (like falling dominoes)
- **Induction makes big problems easier by solving them step by step.**

MATHEMATICAL INDUCTION

- **Mathematical Induction is a special way of proving things.** It has only 2 steps:

Step 1. Verify it is true for the **first one** .

Step 2. Show that if **any one** is true then the **next one** is true .

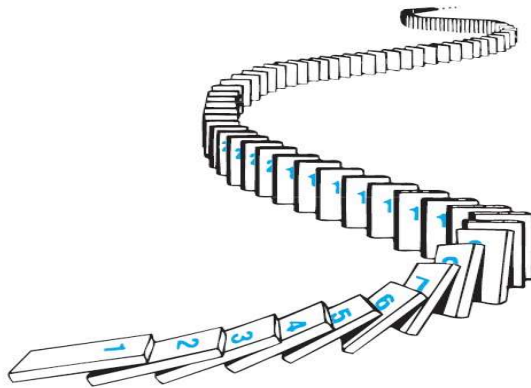
- **Domino effect :**

Step 1. The **first** domino falls

Step 2. When **any** domino falls, the **next** domino falls

So ... **all dominos will fall!**

That's how Mathematical Induction works.



Principal of Mathematical Induction

- ▶ We want to prove a statement $P(n)$ is true for all positive integers.

- ▶ **Step 1: Base Step (Start)**

- ▶ Show it is true for the first value:

$P(1)$ is true

- ▶ **Step 2: Inductive Step (Continue)**

- ▶ Assume it is true for some number k :

*$P(k)$ is true (this is the *inductive hypothesis*)*

- ▶ Then prove it is also true for the next number:

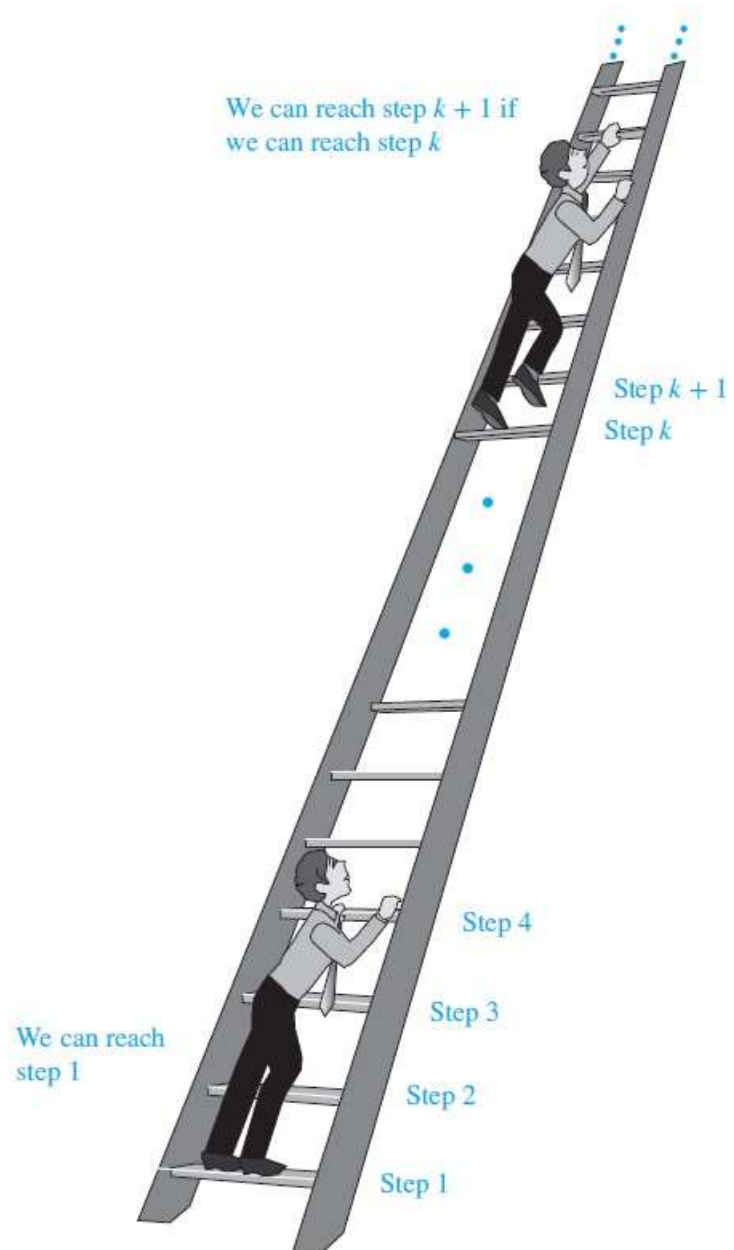
$P(k + 1)$ is true

- ▶ If:

it works for the first number, and
each step leads to the next
then it works for all numbers

**Prove first case + prove next step
works $\rightarrow$ true for all n**

CLIMBING AN INFINITE LADDER.



Proof by INDUCTION

1. Prove the “**Induction Basis**”. That is, prove that the number 1 satisfies the claim. (This step is often easy, but is crucially important, and should never be omitted!)
2. Assume the “**Induction Hypothesis**”. That is, state the assumption that the claim holds for some positive integer K .
3. Prove, using the **Induction Hypothesis**, that the claim holds for $K+1$. The proof should consist of a chain of clear statements, each logically following from the previous ones combined with our shared knowledge base. **The final statement in the chain should state that the claim holds for $k + 1$.**

EXAMPLE 1

1. Use mathematical induction to prove that
$$1 + 2 + 3 + \dots + n = \frac{N(N+1)}{2}$$
for all positive integers $n \geq 1$.

SOLUTION :

Let the statement $P(n)$ be
$$1 + 2 + 3 + \dots + n = \frac{N(N+1)}{2}$$

STEP 1(Induction Basis): We first show that $p(1)$ is true ($n=1$).

$$\text{LHS} = 1$$

$$\text{RHS} = 1(1 + 1) / 2 = 1$$

Both sides of the statement are equal hence **$P(1)$ is true.**

STEP 2: We now assume that $p(k)$ is true ($n = k$)

$$1 + 2 + 3 + \dots + k = \frac{K(K+1)}{2} \longrightarrow \text{Induction hypothesis}$$

Assuming this, we have to now have to proof that the axiom is true for its successor i.e. $p(k+1)$. To do that we must show L.H.S = R.H.s

$$1 + 2 + 3 + \dots + k + (k + 1) = \frac{(k + 1)(k + 2)}{2}$$

1

On the L.H.S of eq 1, $1+2+3+\dots+k$ can be replaced by:

$\frac{K(K+1)}{2}$ using **Induction Hypothesis**. Therefore eq 1 now becomes

$$\Rightarrow \frac{K(K+1)}{2} + (k + 1) = \frac{(K+1)(K+2)}{2}$$

$$\Rightarrow \frac{\{K(K+1)+2(K+1)\}}{2} = \frac{(K+1)(K+2)}{2}$$

$$\Rightarrow \frac{(K+1)(K+2)}{2} = \frac{(K+1)(K+2)}{2}$$

L.H.S = R.H.S

HENCE PROVED

EXAMPLE 2

2. Use mathematical induction to prove that
 $(ab)^n = a^n b^n$.
for every natural number $n \geq 1$.

SOLUTION:

Let the statement $P(n)$ be

$$(ab)^n = a^n b^n$$

STEP 1(Induction Basis):

We first show that $p(1)$ is true ($n=1$).

$$(ab)^1 = a^1 b^1.$$

But $(ab)^1 = ab$; and $a^1 b^1 = ab$.

Both sides of the statement are equal hence **$P(1)$ is true.**

STEP 2: We now assume that $p(k)$ is true ($n = k$)

$(ab)^k = a^k b^k$ $\longrightarrow$ **Induction hypothesis**

Assuming this, we have to now have to proof that this is true for $(k+1)$. To do that we must show L.H.S=R.H.S in eq 1 :

$(ab)^{k+1} = a^{k+1} b^{k+1}$ $\longrightarrow$ **1**

$$a^{k+1} b^{k+1} = a^{k+1} b^{k+1}$$

L.H.S = R.H.S
HENCE PROVED

EXAMPLE 3

Prove using induction that: $1 + 3 + 5 + \dots + (2n-1) = n^2$
for all integers $n \geq 1$.

SOLUTION:

STEP 1: Induction Basis

LHS:

$$1$$

RHS:

$$1^2 = 1$$

LHS = RHS $\rightarrow$ Base case is true

Step 2: Inductive Hypothesis

Assume the statement is true for some k , i.e.,

$$1 + 3 + 5 + \dots + (2k - 1) = k^2$$

EXAMPLE 3

Prove using induction that: $1 + 3 + 5 + \dots + (2n-1) = n^2$
for all integers $n \geq 1$.

SOLUTION:

Step 3: Inductive Step (prove for $n = k+1$)

We need to show:

$$1 + 3 + 5 + \dots + (2k-1) + (2(k+1)-1) = (k+1)^2$$

Start with LHS:

$$= k^2 + (2(k+1)-1)$$

Simplify:

$$\begin{aligned} &= k^2 + (2k + 2 - 1) \\ &= k^2 + (2k + 1) \\ &= (k+1)^2 \end{aligned}$$

This matches RHS

EXERCISE A

1. **Prove that:** $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$
2. **Prove that:** $1+2+2^2 + \dots + 2^n = 2^{n+1} - 1$, for all integers $n \geq 0$
3. **Prove that:** $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, for all integers $n \geq 1$.
4. **PROVE THAT:** $\sum_{i=1}^{n+1} i 2^i = n \cdot 2^{n+2} + 2$, for all integers $n \geq 0$

EXERCISE B

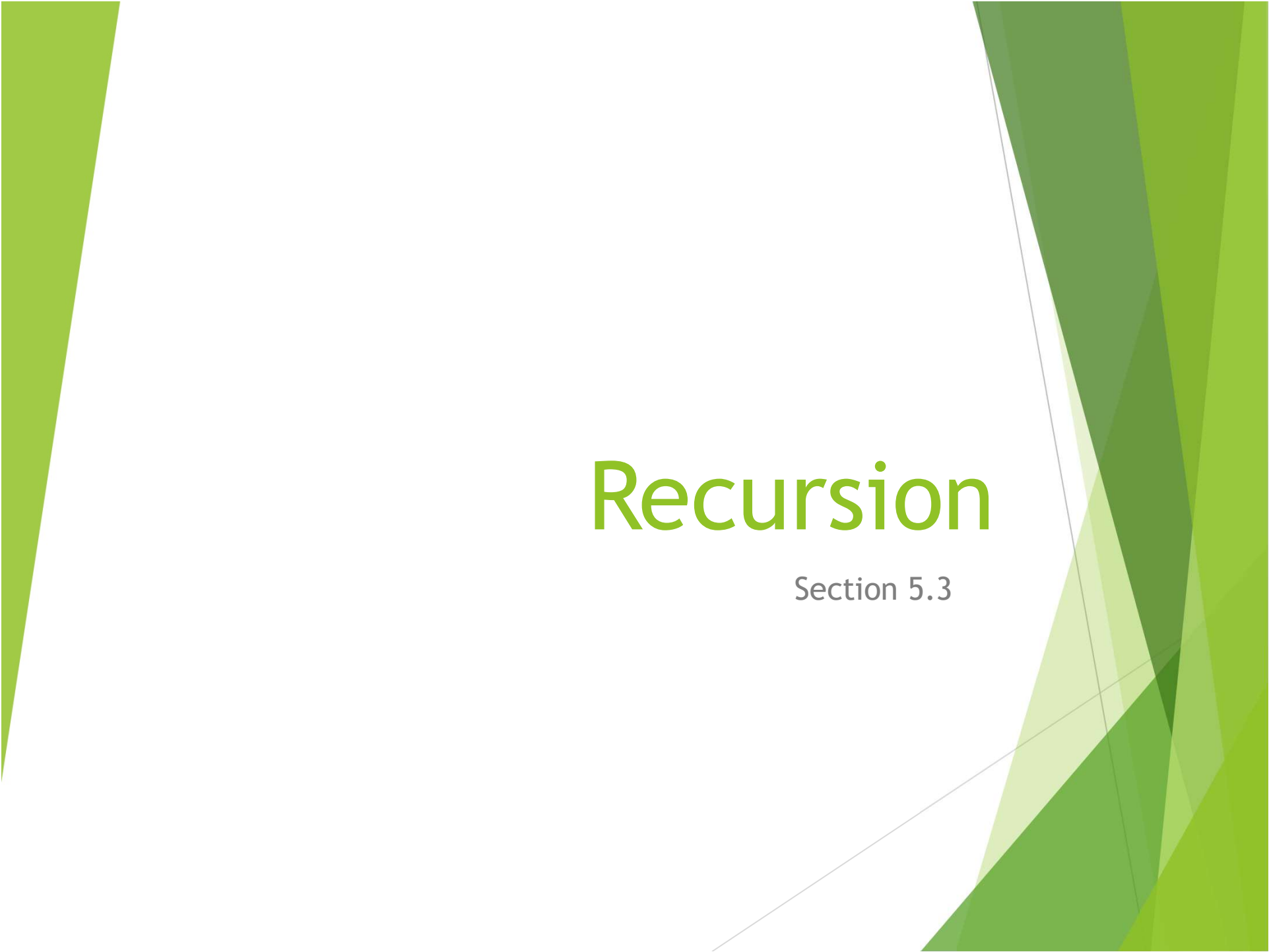
1. Prove by mathematical induction

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1} \quad \text{for all integers } n \geq 1$$

2. Use mathematical induction to prove that

$$\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \quad \text{for all integers } n \geq 2$$



The background features abstract green geometric shapes. On the left, a single green triangle points downwards. On the right, a complex arrangement of overlapping green triangles and polygons in various shades of green (from light lime to dark forest green) creates a layered, architectural effect. A thin, light gray line extends diagonally from the bottom left towards the right, passing behind the green shapes.

Recursion

Section 5.3

What is a Recursive Definition?

- ▶ A **recursive definition** defines something in terms of *smaller versions of itself*.

It always has **two parts**:

- ▶ **Basis Step (Base Case)**

- ▶ This tells you where to start (the simplest value).

- ▶ Example:

$$f(0) = 3$$

- ▶ **Recursive Step**

- ▶ This tells you how to build new values from earlier ones.

- ▶ Example:

$$f(n + 1) = 2f(n) + 3$$

EXAMPLE

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, \dots$, and suppose that $a_0 = 2$.

What are a_1 , a_2 , and a_3 ?

SOLUTION:

We see from the recurrence relation that

$$a_1 = a_0 + 3 = 2 + 3 = 5.$$

It then follows that

$$a_2 = 5 + 3 = 8 \text{ and } a_3 = 8 + 3 = 11.$$

EXAMPLE 6

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$, and suppose that $a_0 = 3$ and $a_1 = 5$.

What are a_2 and a_3 ?

SOLUTION:

We see from the recurrence relation that $a_2 = a_1 - a_0 = 5 - 3 = 2$

and $a_3 = a_2 - a_1 = 2 - 5 = -3$.

We can find a_4 , a_5 , and each successive term in a similar way.

Recursively Defined Functions

Example: Suppose f is defined by:

$$f(0) = 3,$$

$$f(n + 1) = 2f(n) + 3$$

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$

Solution:

- ▶ $f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$
- ▶ $f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$
- ▶ $f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$
- ▶ $f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$

Key Idea:

You *must* go in order.

You can't jump to $f(4)$ without knowing $f(3)$.

Recursively Defined Functions

Example: Give a recursive definition of the factorial function $n!$

Suppose f is defined by:

$$f(0) = 1$$

$$f(n + 1) = (n + 1) \cdot f(n)$$

Find $f(1)$, $f(2)$, $f(3)$, $f(4)$

Solution:

$$f(0) = 1$$

$$f(1) = 1 \times f(0) = 1$$

$$f(2) = 2 \times f(1) = 2$$

$$f(3) = 3 \times f(2) = 6$$

$$f(4) = 4 \times f(3) = 24$$

Factorial means:

$$n! = n \times (n-1) \times (n-2) \times \dots \times 1$$

Recursive definition just breaks that into steps.

Recursively Defined Functions

Example: The Fibonacci numbers are defined as follows:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2}$$

Find f_2, f_3, f_4, f_5 .

Solution:

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

EXERCISE A

1. Find the first four terms of the following recursively defined sequence.

$$b_1 = 2, \quad b_k = b_{k-1} + 2 \cdot k, \quad \text{for all integers } k \geq 2$$

2. Find the first five terms of the following recursively defined sequence.

$$t_0 = -1, \quad t_1 = 1, \quad t_k = t_{k-1} + 2 \cdot t_{k-2}, \quad \text{for all integers } k \geq 2$$

3. Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$ if $f(n)$ is defined recursively by $f(0) = 1$ and for $n = 0, 1, 2, \dots$

a) $f(n+1) = f(n) + 2.$

b) $f(n+1) = 3f(n).$

c) $f(n+1) = 2f(n).$

d) $f(n+1) = f(n)^2 + f(n) + 1.$